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CSA 0315

DATA STRUCTURES

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1) Write a c-program to count Number of odd & even elements in an array of N-elements.

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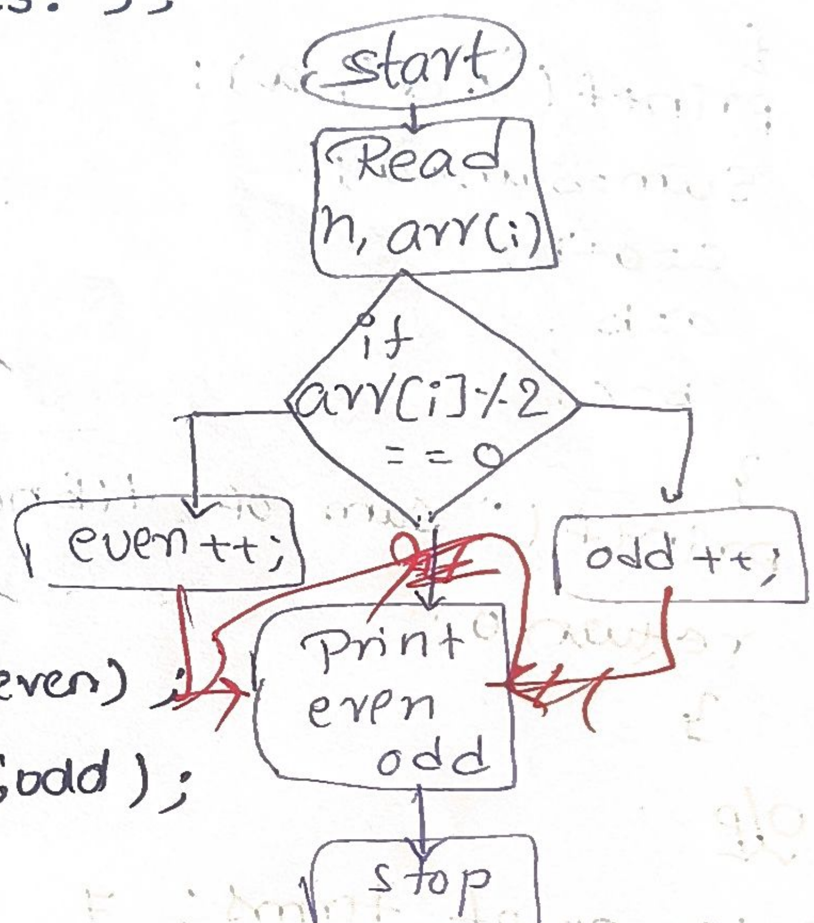
Code:

```
#include <stdio.h>
int main()
{
int n,i,even=0,odd=0;
printf("enter number of elements:");
scanf("%d", &n);
 for(i=1 ;i<n;i++)
 {
 scanf("%d" ,&a[i]);
 if(arr[i]%2==0)
  even ++;
 else
  odd++;
 }
printf("even elements= %d\n",even);
printf("odd elements= %d\n",odd);

return0;
 }
```



O/P

enter number of elements: 5

10 15 16 8 3

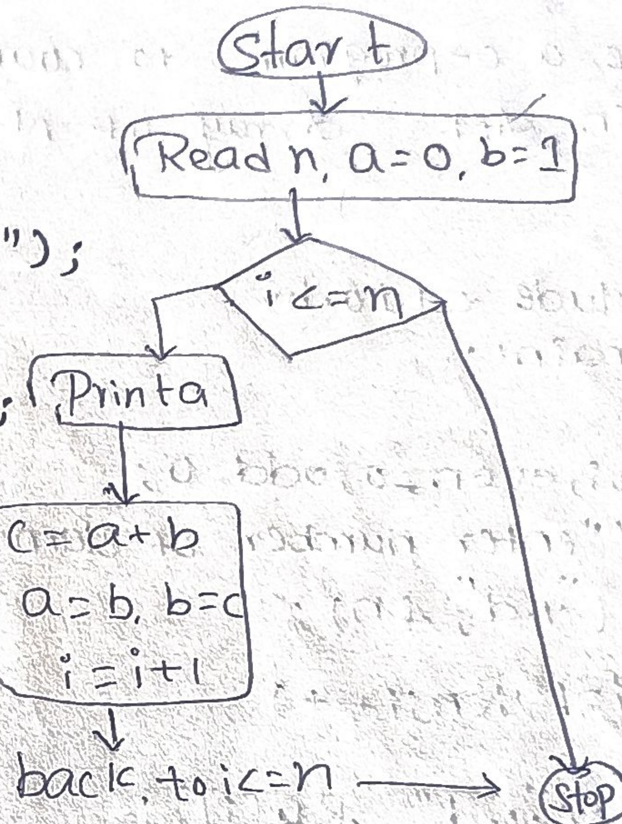
even elements: 3

odd elements: 2

2) write a c program to print fibbinocci & sum of fibbocci.

Code:

```
#include <stdio.h>
int main()
{
int n,i,a=0,b=1,c,sum=0;
printf("enter no.of Terms:");
Scanf("%d",&n)
printf("fibonacci series:");
for(i=1;i<n;i++)
{
printf("%d",a);
Sum=sum+a;
c=a+b;
a=b;
b=c;
}
printf("Sum of fibnocci series:%d",Sum);
return 0;
}
```



O/P

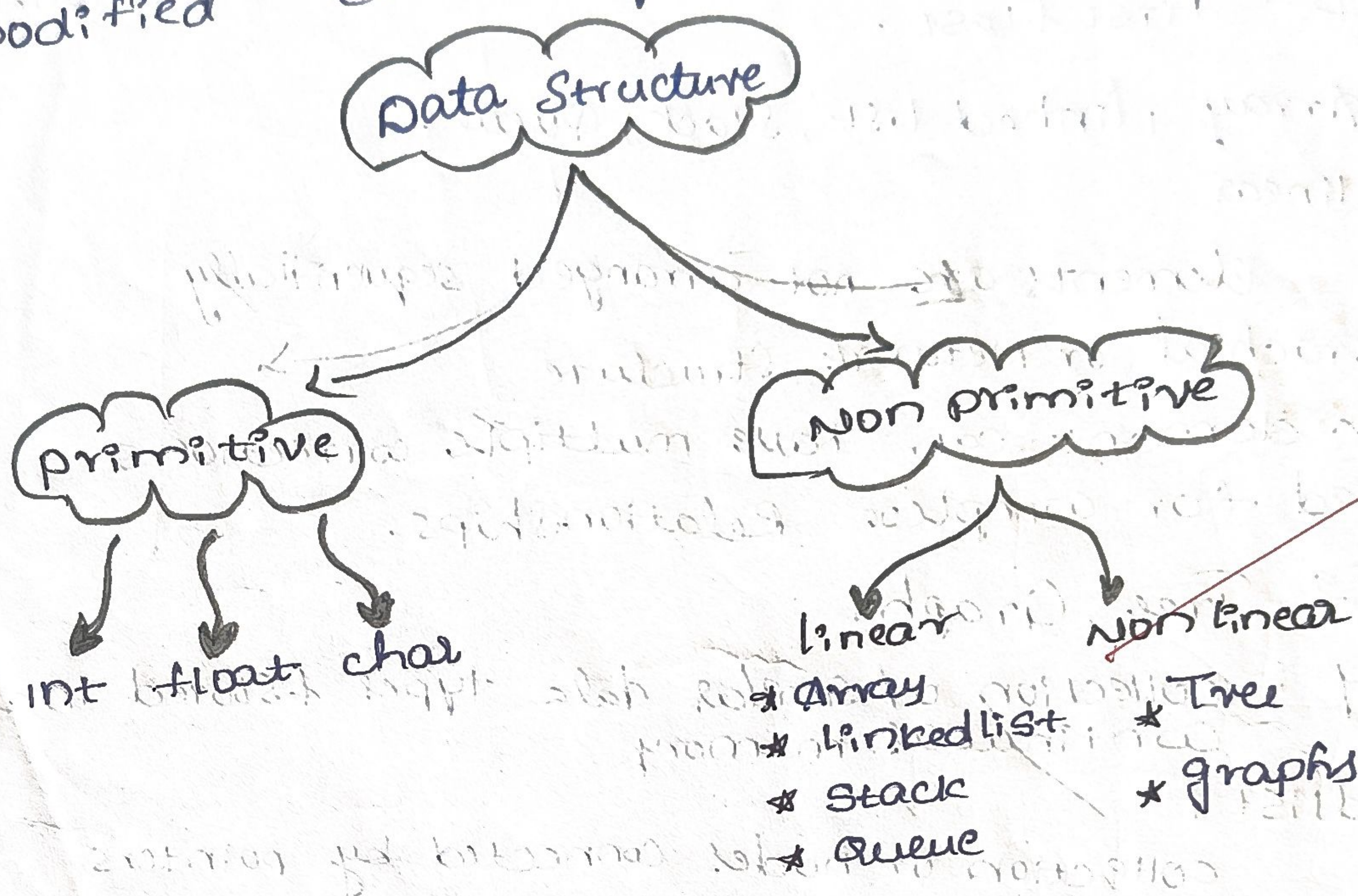
enter no.of terms: 7

fibnacci Series: 0 1 1 2 3 5 8

Sum of fibonacci series: 20

Write a concept map of data structures & classification.

A data structure is a way of organizing storing and managing data so that it can be accessed & modified efficiently.



primitive data structure:

There are the basic types which are provided by a programming language

eg: int, float, char, double, boolean

Non primitive data structure:

These are deserved from primitive data types and can store multiple values.

linear

elements are arranged are after another in a sequence.

* sequential arrangement
* easy Transverse
* Each element has a one predecessor & successor except first & last.

Eg Array, linked list, Stack, Queue.

Non linear

elements are not arranged sequentially

→ Hierachial or Network Structure
→ one element can have multiple corrections.
→ used for complex Relationships.

Eg Tree, Graphs

Array : collection of similar data types & stored in a Continuous memory

linkedlist :
collection or nodes connected by pointers

stack : follows LIFO (last in first out) it operates push & pop etc

Queue :
Follows FIFO

Tree :
Hierachial data structure

Graph:
Collection of nodes & edges.